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Purpose/Policy:

This CFT Filler and Seamer standard operating procedure has been created to assist operators with setup, SIP/CIP, changeovers, basic cleaning, and troubleshooting.

Scope:

The CFT filler is mainly comprised of 32 fill valves on a rotating column. Cans are fed into the individual filling valves by a worm wheel and then a star wheel. The speed at which the filler runs is dependent on the packaging size we are running. Packageable liquid is stored in a “bowl” behind the filling valves and is kept at a constant pressure by a CO2 supply.

After the cans are filled, they travel along a short transfer rail to the seamer where a lid is dropped onto them and sealed as they rotate. Before cans reach the seamer, they pass under the undercover gasser which puts a layer of CO2 on top of the cane before the lid drops. The seamer is a 6 head seamer unit. Lids are fed into the seamer by a lid slide that is constantly being loaded by a filler operator.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- E-Stop – Emergency Stop button
- SIP – Sanitization in Process
- CIP – Cleaning in Process
- DAW – De-Aerated Water (D.O. water)
- D.O. – Dissolved Oxygen
- CO2 – Carbon Dioxide
- PAA – Peroxyacetic Acid, sanitizer (Rochester Midland)
- F400 – Caustic cleaning solution (Rochester Midland)
- Perafoam – Foaming additive for PAA foaming (Rochester Midland)
- QC – Quality Control
- HLC5000 – Caustic cleaner for CIP use (Rochester Midland)
- HDA2500 – Acid cleaner for CIP use (Rochester Midland)
- CFT Filler – The 32-valve rotary can filler (CFT Filler and Seamer 3000/6)
- Seamer – The 6-head unit that drops a lid onto each can and seals it
- Undercover Gasser – Puts a layer of CO2 on top of the can before the lid drops
- Bowl – Tank behind the filling valves that holds packageable liquid under CO2 pressure

- Brink – Tank that holds cleaning or sanitizing media (caustic, acid, hot water, PAA) for the filling units

Authorities/Responsibilities:

“This SOP can be completed by a Lead level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to operate the “CFT Filler” and trouble shoot any issues that may arise.”

Resources and Materials:

Operations:

- Lids (CDL11, Gold CDL 11, Cider Ends)
- Liquid Product
- DAW (DO Water)
- CO2/Nitrogen
- Air Tool Oil
- Din fitting wrench
- Autoclaved Pinpoint Filter
- Raw Cans (12oz, 16oz, 19.2oz)

Changeovers:

- 16oz/19.2oz Star wheel
- 16oz Guide Rail
- 19.2oz Guide Rail
- 19.2oz Seamer Star Wheel
- 13mm wrench
- 6mm hex wrench
- 8mm hex wrench
- 10mm hex wrench
- 12oz pin-height guide and shims
- 16oz pin-height guide and shims
- 19.2oz pin-height guide

Basic Cleaning:

- CIP Cups
- F400 Foamer
    - F400 Chemical

- PAA Foamer
    - Oxystrike Chemical and Perafoam foaming agent.
- Water (Mild temp)

Deep Clean:

- See the “Deep Cleaning” Section of this SOP Document.

Troubleshooting:

- Cup Adjustment tool.
- Mouth Seal Removal Tool.

PPE Required:

- For SIP and CIP Brink – Green Rubber Gloves, Face Shield and Safety Glasses.
- For Foaming – Safety Glasses, Face Shield and Nitrate Gloves
- For CO2 process – Hearing Protection.

Procedures:

Setting up the CFT Filler for SIP:

Filling the Lifter Oil:

1. Prior to proceeding, ensure the air to the CFT Filler is in the “Off” position.
2. Open the Lifter Oil Cabinet located behind the Filter Skid.
3. Loosen the right-side metal reservoir.
4. Fill with Air tool oil.
5. Hand tighten the metal reservoir.
6. Proceed to tun the air to the “On” position for the CFT Filler.

Putting SIP/CIP Cups on CFT filler:

Cups may already be depending on the last process, but if they are not, please use the following steps to complete.

1. Undo the Emergency Stops and ensure the Filler doors are closed.
2. Ensure the CFT Filler, “Filler in CIP” button is “Enabled.”
    2.1 To check this, do the following:
        2.1.1 Select the “Beaker” button from the HMI Panel.
        2.1.2 If the “Filler in CIP” is red, select the “Enable/Disable” Button till it turns Green.

3. Put the filler into jog mode using the key next to the alarm reset button below the HMI panel view.
4. Start putting on the CIP cups and jog the filler as needed to ensure all valves have a CIP cup installed.

Note: Pay attention as If a cup falls off during the jogging step it can cause massive damage to the filler.

5. Now that all cups are installed, open inlet of 1µ (Micron) filter housing and the outlet of the 1µ (Micron) filter housing as this housing is used for PAA.
6. Connect a hot water hose to the 20µ filter housing.
    6.1 Open the 20µ inlet and slightly open the 20µ back drain to let hot water rinse out for 20-minutes.
    6.2 Slightly open the 20µ filter housing top valve to let hot water rinse out for 20-minutes.

Note: This can continue to run as you proceed through the next steps.

Setting up the SIP/CIP Brink:

1. Head to the Cellar
2. Connect the “Can Line Product Hose” to the CIP Brink.
3. Connect the Cold-Water Hose (located behind the brink) to Valve R1.
4. Open the Can-Line Product Hose, Valve R1 and turn on the Cold-Water Hose.
5. Open Valve R2 (The Can Line Flow Plate Valve)
6. If using hard pipe, use the following steps:
    6.1 Open R4 (Drain) halfway.
    6.2 Close R3 (Brink Return)
    6.3 Open R5 (Hard Pipe Return).
    6.4 Ensure the Red and Blue Line Hard Pipes are in the open positions in both the new and old Cellar.
    6.5 Open R6 (Brink Drain)
    6.6 Close R7 (Brink to Product line)
7. For non-hard pipes, use the following steps:
    7.1 Open R4 (Drain).
    7.2 Open R3 (Brink Return) halfway.
    7.3 Ensure R5 (Hard Pipe Return) is closed.
    7.4 Open R6 (Brink Drain)
    7.5 Close R7 (Brink to Product line).
8. Return to CFT Filler.

Starting the CFT Filler SIP Cycle:

1. On the HMI panel select the home screen (House button).
2. Using the" "rolodex"" button, tab up one page.
3. Turn the following sections to the auto position:
   - 3.1 Opening Filling Valves.
   - 3.2 VP25 Product Infeed.
   - 3.3 PO2 Infeed Product Pump.
   - 3.4 VM2 Counterpressure Mod.
   - 3.5 VM3 Breather Mod.
4. Using the "rolodex" button, tab up one page.
5. Turn on the Machine Light.
6. Select the "Beaker" to get to the SIP/CIP screen.
7. Press "Start CIP"
8. "rolodex" up one using the "rolodex" button.
9. Press the "Force" button next to the "Water Cycle On" label.

Note: This calls the water from the Cellar to the CFT Filler.

10. *Ensure the bowl fills to 95% and then the cycle will start.*
11. *Return to the Brink in the Cellar.*
12. *Ensure clear cool water is flowing (should be returning into brink and drain).*
    - 12.1 *If the water is hot from the night before, continue to let the cycle run until the water is cold.*
13. *Wear the Required PPE, proceed to the chemical room to grab Oxystrike (PAA).*
    - 13.1 *For "Hard-Pipes", collect 2000ml of Oxystrike*
    - 13.2 *For "Non-Hard-Pipes" collect 1500ml of Oxystrike.*
14. *Return to the brink with chemical to start the filling process.*
15. *To fill the brink, use the following processes:*
    - 15.1 *"Hard-Pipes":*
      - 15.1.1 *Close valve R6.*
      - 15.1.2 *Close valve R4.*
      - 15.1.3 *Water will start to fill the brink.*
      - 15.1.4 *Add the 2000ml of Oxystrike to the brink.*
      - 15.1.5 *Fill the brink up to the welded bolt (halfway up brink).*
      - 15.1.6 *Once brink is "full," open valve R7 and close valve R1.*
      - 15.1.7 *This will start your 50-minute cycle.*
    - 15.2 *"Non-Hard-Pipes":*
      - 15.2.1 *Close valve R6.*
      - 15.2.2 *Close valve R4.*

15.2.3 *Water will start to fill the brink.*
15.2.4 *Add the 1500ml of Oxystrike to the brink.*
15.2.5 *Fill the brink up to the welded bolt (halfway up brink).*
15.2.6 *Once brink is "full," open valve R7 and close valve R1.*
15.2.7 *This will start your 50-minute cycle.*

16. *After 10 minutes have elapsed, we need to bleed the 1µ filter housing and pinpoint carbonator (only on first run of week).*
    16.1 *Open the front of the filter housing for a moment to let water flow out, creating contact with the pipe.*
    16.2 *Open the dirty and clean drains (bottom of the filter housing) and let water flow for a moment.*
    16.3 *Open the outlet of the filter housing and let water flow for a moment.*
    16.4 *Open the top valve of the filter housing and let water flow for a moment.*

Foaming/Ending CFT Filler SIP cycle:

Foaming the CFT Filler/Seamer and Conveyor Belts

Notes: Safety glasses, Rubber gloves and face shield must be worn while foaming.

1. Grab the Oxystrike foaming unit.
2. Remove cap and fill the unit with cold water up till the 5-gallon mark.
3. While wearing proper PPE, collect 100ml of Oxystrike and add to the Oxystrike foaming unit.
4. Pour in 3oz of Perafoam into the Oxystrike foaming unit.
5. Add cap back to foaming unit.
6. Connect the air line to the seamer air line drop and the other end to the foaming unit.
7. Foam the CFT Filler, Seamer and Conveyors per training instructor.

Ending the CFT Filler SIP cycle:

1. After the 50-minutes have elapsed, it is time to end the SIP cycle.
2. On the CIP screen on the HMI panel, press "Force" next to the "Empty Cycle On (Step 9)" label.

Note: This will start the draining cycle back to the brink.

3. Head to the brink in the Cellar and open valve R4 and valve R6.
4. Close valve R7 and close "Can Line Product Hose" valve.

Hooking up to DAW Water:

1. After the SIP cycle has ended, the next steps will be for purging the CFT Filler.

1.1 Hooking up to DAW.

1.1.1 After the product line has been closed from the brink, remove the product line, and attach to DAW drop (for non-hard-pipes) and attach to the “Red” line tree (for hard-pipes)

Note: For hard pipes, the DAW drop is in the new Cellar. The blue line needs to be removed and attached to the DAW drop in order to send DAW water from the new Cellar to the filler.

Note: Before running DAW, please verify with a Cellar operator that DAW can be used.

1.1.2 *Once the Cellar has approved DAW use, we can open the DAW supply and the Can Line Product valve.*

1.1.3 *Return to the CFT Filler.*

Removing SIP/CIP Cups:

1. *While the CFT Filler is still in CIP Enabled mode, we will need to remove all the “CIP Cups” that we placed on the filler earlier in this SOP.*
   1.1 *Using the jog wand, jog the filler and remove all cups.*

Purging CFT Filler with DAW:

1. *As the line is already opened for DAW water, we will need to purge the filter housings and pinpoint with DAW water.*
2. *Use the following steps to rinse out the filter housings and pinpoint.*
   2.1 *Open the front of the 1µ filter housing for a moment to let water flow out to rinse Oxystrike out.*
   2.2 *Open the dirty and clean drains (bottom of the 1µ filter housing) and let water flow to rinse Oxystrike out.*
   2.3 *Open the outlet of the1µ filter housing and let water flow to rinse Oxystrike out.*
   2.4 *Open the top valve of the filter housing and let water flow to rinse Oxystrike out.*
   2.5 *Open the inlet to the 20µ (Micron) filter housing and then close the inlet to the 1µ housing.*
   2.6 *Open the outlet to the 20µ filter housing and then close the outlet to the 1µ filter housing.*
   2.7 *Proceed to rinse the 20µ house as we did in the steps above.*
   2.8 *Open the top and bottom drains of the Pinpoint Carb unit to let DAW flow for a moment.*
3. *Using the following steps to rinse the CFT Filler with DAW.*
   3.1 *On Filler HMI, select the “Beaker” and disable the “Filler in CIP.”*
   3.2 *Turn the key on the Filler HMI to “Auto.”*

3.3 *Hit the reset button on the Filler HMI.*
3.4 *Press the Green "Start" Button.*
3.5 *From the Seamer cabinet, open the water valve.*
3.6 *Select the Home Screen Icon and "rolodex" up three tabs and press start washing to engage the Seamer Wash.*
3.7 *To start flushing DAW through the filler, use the following steps:*
- 3.7.1 *From the home screen, "rolodex" up one tab.*
- 3.7.2 *Put the following buttons into "Manual":*
  - 3.7.2.1 *Opening Filling Valves.*
  - 3.7.2.2 *VP25 Product Infeed.*
  - 3.7.2.3 *PO2 Infeed Product Pump.*
  - 3.7.2.4 *VM3 Breather Mod.*
- 3.7.3 *Put the following button into the "Off" position:*
  - 3.7.3.1 *VM2 Counterpressure Mod.*
- 3.7.4 *This will allow DAW to flow through the filling valves and onto the filler deck, rinsing the Oxystrike chemical from the valves.*
- 3.7.5 *Leave the water cascading for two full rotations.*
- 3.7.6 *Proceed to turn all the buttons to the off position.*
- 3.7.7 *Next, use the following steps to fill the bowl and then dump manually:*
  - 3.7.7.1 *Leave "Opening Filling Valves" in the off position.*
  - 3.7.7.2 *Turn "VP25 Product infeed, PO2 Infeed Product Pump and VM3 Breather Mod" to the "Man" position.*
  - 3.7.7.3 *Once the bowl reaches 100%, proceed to dump the bowl using the following steps:*
  - 3.7.7.4 *Turn off the ""VP25 Product Infeed " and PO2 Infeed Product Pump".*
  - 3.7.7.5 *Turn the "Opening Filling Valves" to the "MAN" position.*
  - 3.7.7.6 *This will drain the bowl.*
- 3.7.8 *Repeat steps 3.7.7.1- 3.7.7.4 one more time.*
- 3.7.9 *Proceed to turn all buttons to the "Off" position except" VM3 Breather Mod and VM2 Counterpressure Mod." These two should be switch to the "auto" position.*

3.8 *This completes the DAW flush of the CFT Filler.*

Purging CFT Filler with CO2:

Note: This section will only be relevant for the first Beer SIP of the week or after a night in which the pinpoint carbonator is used.

Note: This section requires hearing protection.

1. *Return to the Cellar where the "Can Line Product Hose" is hook up to DAW and shut off the DAW valve and product line valve.*
2. *Connect the "High Pressure CO2 Line" to the "Can Line Product Hose" (There is one in the new Cellar and one in the old Cellar which will be used depending on where your product line is)*
3. *Turn on the CO2 valve and open the "Can Line Product Hose" valve.*
4. *Return to the CFT Filler.*
5. *Open the front of the 20µ filter housing for a moment to let water flow out as CO2 is pushing it.*
6. *Open the dirty and clean drains (bottom of the 20µ filter housing) and let water flow out while CO2 is purging.*
7. *Open the outlet of the20µ filter housing and let water flow out while CO2 is purging.*
8. *Open the top valve of the filter housing and let water flow out while CO2 is purging.*
9. *After the filter housing seems to have most of the water blown out by the CO2 purge, close each valve one at a time. This will allow pressure to build in the filter housing.*
10. *Open the inlet valve to blow out built up pressure and then close the inlet valve.*
11. *Once pressure has built back up, open the outlet valve to release the pressure.*
12. *Repeat until most of the water is out of the filter housing.*
13. *At the Filler HMI use the following steps to blow out any remaining DAW with CO2:*
    - 13.1 *From the home screen, "rolodex" up one screen.*
    - 13.2 *Ensure the following controls are in the off position:*
        - 13.2.1 *Filling Valves.*
        - 13.2.2 *VP25 Product infeed.*
        - 13.2.3 *P02 Infeed Product Pump.*
    - 13.3 *Ensure the VM2 Counterpressure Mod and VM3 Breather Mod are in the auto position.*
    - 13.4 *When the bowl is at setpoint for pressure, open the Filling Valves to the "man" position.*
    - 13.5 *When the bowl pressure drops to about 3-2psi, turn VM2 Counterpressure Mod to "man".*
    - 13.6 *Once the bowl blows out CO2, close the "Filling Valves" and return "VM2 Counterpressure Mod" to "auto". The bowl will begin to build up pressure to the setpoint again.*
    - 13.7 *Once the bowl reaches the setpoint again, repeat steps 13.4-13.6 three times.*
    - 13.8 *At this point, the HMI should read everything off except VM2 and VM3 which should be left in "auto".*
14. *After the bowl has been blown out, head over to the pinpoint carbonator.*
15. *Open the drain port on the pinpoint filter housing.*

16. *Once water has finished being blown out, close the drain port.*
17. *Close "CIP Supply" and "To Filler PP Carbonator."*
18. *Open the drain port again on the pinpoint to relieve any pressure.*
19. *Open housing and add autoclaved filter to pinpoint filter.*
20. *Close filter housing.*
21. *Change the swing panel position using the din fitting wrench so the panel is hooked up to "CO2 Supply and "To Filler PP Carbonator."*
22. *Open the valves to "CO2 Supply" and "To Filler PP Carbonator."*
23. *Open the drain port of the pinpoint housing.*
24. *Open Pinpoint cabinet and turn CO2 supply knob till you have co2 flow coming out of the pinpoint drain port.*
25. *Close the pinpoint drain port and turn the CO2 supply knob to zero.*
26. *Close "CO2 Supply" and "To Filler PP Carbonator" valves.*
27. *Close the blue handled valve behind the pinpoint carbonator.*
28. *Return to the Cellar, shut off CO2 supply, and then close the ""Can Line Product Hose" Valve."*

Pulling Product:

1. After the packaging split has been sent out and the tank has been signed off, it is time to pull the product.
2. In the Cellar, we need to connect our "Can Line Product Hose" to the product we will be running. This can include directly to the tank in the old Cellar or hooking the hose up to the red line drop tree and using the blue line hose in the new Cellar to connect to the tank.
3. Before opening product valves, hook up CO2 or Nitrogen (product depending) to the vent arm.
4. Once CO2/Nitrogen has been turned on and set to correct pressure, open the product hose valve and the product valve in the tank.
5. Return to the CFT Filler.
6. From the HMI screen, turn PO2 Infeed Product Pump and VP25 Product Infeed

Note: This section requires hearing protection for purging product through filter housings.

7. Open the inlet to the 20µ (Micron) filter housing and product will eventually flow out. Once this happens close inlet valve.
8. Purge all outlets till beer has flowed through.
9. From the Filler HMI, wait till the bowl hits set point.
10. Once the bowl is at setpoint, close "PO2 Product infeed" and "VP25 Product infeed."
11. Open the Filling Valves to the "man" position. This will cause the bowl to drain once the pressure is relieved.

12. After the dump has been completed, return everything back to the “auto” position.

Note: This step may need to be repeated to qualify in spec.

- 12.1 During this time, please ensure that lids have been loaded on the lid slide and into the Seamer.
- 12.2 Ensure cans have been brought up (See Depal SOP CL01) and have been properly date coded (See Date Code SOP and Can Coder SOP CL04)
- 12.3 Place the stop bar in the line before the warmer to prevent cans from going into it.

13. From the HMI home screen, open the “Container Gate” and run 8-10 cans through the filler and then close the “Container Gate”.
    - 13.1 The first six cans will be used for seam checks.
    - 13.2 Looking into the Seamer, the base has numbers 1-6 on them. The can that goes in first will determine the order in which you need to grab them.
    - 13.3 The remaining cans will be used for QC checks and taste/visual approval.
14. Once the QC check have been completed and verified by two operators, we are now able to start production.
15. Remove the bar and then return the filler HMI and open the “Container Gate.” You have now started your production run.

End of Run DO/HOT Push:

1. The production run will end when we either hit our scheduled number or we have kicked the Bright Beer Tank depending. Grab the last can through the filler and save it for a pre-post push comparison.
2. On the filler HMI home screen, close the “Container Gate,” shut off “VP25 Product Infeed” and the “PO2 Infeed Product Pump.”

Note: We would need to shut off glycol if we kicked the tank. There will be training on how to do this and not in this SOP.

3. In the Cellar, disconnect the product line from the tank and hook up to a hot water line or DAW drop (depending).
4. Open the hot water valve or DAW valve to begin flow.
5. Return to the filler.
6. On the HMI screen, turn the “PO2 Infeed Product Pump” and “VP25 Product Infeed” to “auto” to start filling the bowl again.
7. Once the bowl has reached its set point, open the “Container Gate”, and continue to run cans.
8. We will need to keep our eyes on the sight glass to watch for color change (water mixing with the beer) and when we see that, we will shut off “VP25 Product Infeed “ and run cans

through the filler until the bowl reaches 20% and then we will close the “Container Gate”. Grab the last can through the filler.

9. Pour and compare the pre and post push samples. If there are any concerns, please notify QC.

Dumping the line:

1. From the CFT Filler HMI, “rolodex” up one page.
2. Place the Filling Valves into “man,” VM2 Counterpressure Mod to the off position and VM3 Breather Mod to the “man” position.
3. Place “PO2 Product Pump “to the off position to dump the remaining beer in the bowl.
4. Once all the remaining beer has been depleted, turn on everything but the Filling valves back to the “auto” position.
5. Head over to the filter housing and rinse out housing until all beer has been pushed out by opening all the valves.
6. Back on the Filler HMI, set the “PO2 Product Pump” and “VP25 Product Infeed “to manual. This will allow the bowl to fill up to 100%.
7. Shut off the “PO2 Product Pump” and “VP25 Product Infeed “and set the Filling Valves to “MAN” to flush the filler.
8. Repeat steps 4-7 as necessary to rinse bowl of product.
9. Once the bowl has been rinsed, turn all controls to the “Off” position.

Note: If proceeding to the next product, repeat the “Pulling Product” steps.

Note: On the last run of the week, perform a Seamer wash.

Setting up the CFT Filler for CIP:

1. Once the bowl has been cleared and the hot flush has been completed, we will need to put cups on the CFT Filler.
2. Ensure the CFT Filler, “Filler in CIP” button is “Enabled.”
3. To check this, do the following:
4. Select the “Beaker” button from the HMI Panel.
5. If the “Filler in CIP” is red, select the “Enable/Disable” Button till it turns Green.
6. Put the filler into jog mode using the key next to the alarm reset button below the HMI panel view.
7. Start putting on the CIP cups and jog the filler as needed to ensure all valves have a CIP cup installed.

Note: Pay attention as If a cup falls off during the jogging step it can cause massive damage to the filler.

Setting up the SIP/CIP Brink:

1. Head to the Cellar
2. Connect the “Can Line Product Hose” to the CIP Brink.
3. Connect the Hot-Water Hose (located behind the brink) to Valve R1.
4. Open the Can-Line Product Hose, Valve R1 and turn on the Hot-Water Hose.
5. Open Valve R2 (The Can Line Flow Plate Valve)
6. If using hard pipe, use the following steps:
    - 6.1 Open R4 (Drain) halfway.
    - 6.2 Close R3 (Brink Return)
    - 6.3 Open R5 (Hard Pipe Return).
    - 6.4 Ensure the Red and Blue Line Hard Pipes are in the open positions in both the new and old Cellar.
    - 6.5 Open R6 (Brink Drain)
    - 6.6 Close R7 (Brink to Product line)
7. For non-hard pipes, use the following steps:
    - 7.1 Open R4 (Drain).
    - 7.2 Open R3 (Brink Return) halfway.
    - 7.3 Ensure R5 (Hard Pipe Return) is closed.
    - 7.4 Open R6 (Brink Drain)
    - 7.5 Close R7 (Brink to Product line).
8. Return to CFT Filler.

Starting the CFT Filler CIP Cycle:

1. On the HMI panel select the home screen (House button).
2. Using the “rolodex” button, tab up one page.
3. Turn the following sections to the “auto” position:
    - 3.1 Opening Filling Valves.
    - 3.2 VP25 Product Infeed.
    - 3.3 PO2 Infeed Product Pump.
    - 3.4 VM2 Counterpressure Mod.
    - 3.5 VM3 Breather Mod.
4. Using the “rolodex” button, tab up one page.
5. Turn on the Machine Light.
6. Select the “Beaker” to get to the SIP/CIP screen.
7. Press “Start CIP”
8. “rolodex” up one using the “rolodex” button.
9. Press the “Force” button next to the “Water Cycle On” label.

Note: This calls the water from the Cellar to the CFT Filler.

10. *Ensure the bowl fills to 95% and then the cycle will start.*
11. *Return to the Brink in the Cellar.*
12. *Ensure clear hot water is flowing (should be returning into brink and drain).*
13. *Wear the Required PPE, proceed to the chemical room to grab HLC5000 (Caustic).*
    - 13.1 *For "Hard-Pipes," collect 2.7 gallons of Caustic.*
    - 13.2 *For "Non-Hard-Pipes," collect 1.7 gallons of Caustic.*
14. *Return to the brink with chemical to start the filling process.*
15. *To fill the brink, use the following processes:*
    - 15.1 *"Hard-Pipes":*
      - 15.1.1 *Close valve R6.*
      - 15.1.2 *Close valve R5 for now.*
      - 15.1.3 *Using the cold-water hose, fill the brink halfway from the welded pin.*
      - 15.1.4 *Open Valve R5*
      - 15.1.5 *Close valve R4.*
      - 15.1.6 *Hot Water will start to fill the brink.*
      - 15.1.7 *Add 2.8 gallons of Caustic to the brink.*
      - 15.1.8 *Fill the brink up to the welded bolt (halfway up brink).*
      - 15.1.9 *Once brink is "full," open valve R7 and close valve R1.*
      - 15.1.10 *This will start your 50-minute cycle.*
    - 15.2 *"Non-Hard-Pipes":*
      - 15.2.1 *Close valve R6.*
      - 15.2.2 *Close valve R3.*
      - 15.2.3 *Using the cold-water hose, fill the brink halfway from the welded pin.*
      - 15.2.4 *Open valve R3.*
      - 15.2.5 *Close valve R4.*
      - 15.2.6 *Water will start to fill the brink.*
      - 15.2.7 *Add 1.7 gallons of Caustic to the brink.*
      - 15.2.8 *Fill the brink up to the welded bolt (halfway up brink).*
      - 15.2.9 *Once brink is "full," open valve R7 and close valve R1.*
      - 15.2.10 *This will start your 50-minute cycle.*
16. *After 10 minutes have elapsed, we need to bleed the 20µ filter housing and pinpoint carbonator (only on last run of week).*
    - 16.1 *Open the front of the filter housing for a moment to let water flow out, creating contact with the pipe.*
    - 16.2 *Open the dirty and clean drains (bottom of the filter housing) and let water flow for a moment.*
    - 16.3 *Open the outlet of the filter housing and let water flow for a moment.*

16.4 *Open the top valve of the filter housing and let water flow for a moment.*

Foaming/Acid Cycle/Ending CFT Filler CIP cycle:

Foaming the CFT Filler/Seamer and Conveyor Belts

Notes: Safety glasses and face shield must be worn while foaming. Nitrate gloves are recommended.

1. Grab the F400 foaming unit.
2. Remove cap and fill the unit with warm water up till the 5-gallon mark.
3. While wearing proper PPE, collect 4oz (118ml) of F400 and add to the F400 foaming unit.
4. Add cap back to foaming unit.
5. Connect the airline to the seamer airline drop and the other end to the foaming unit.
6. Foam the CFT Filler, Seamer and Conveyors per training instructor.

Performing Last Run Acid Cycle:

Note: The acid cycle is performed after the last production run of the week.

1. After the 50-minute Caustic cycle has been completed, return to the CIP brink.
2. Remove the Hot-Water Hose from valve R1 and replace it with the cold-water hose.
3. Open valve R1.
4. Close valve R7.
5. Open valve R6.
6. Open valve R4 halfway.
7. Let cold-water flush for 15-minutes or until all hot water has been cleared from the line.
8. Once cold water is present and all hot water/caustic has been flushed from the system, it is time to begin the Acid cycle.
9. *Wear the Required PPE, proceed to the chemical room to grab HDA2500 (Acid).*
   - 9.1 *For "Hard-Pipes," collect four gallons of Acid.*
   - 9.2 *For "Non-Hard-Pipes" collect three gallons of Acid.*
10. *Return to the brink with chemical to start the filling process.*
11. *To fill the brink, use the following processes:*
    - 11.1 *"Hard-Pipes":*
      - 11.1.1 *Close valve R6.*
      - 11.1.2 *Close valve R4.*
      - 11.1.3 *Cold-Water will start to fill the brink.*
      - 11.1.4 *Add 4 gallons of Acid to the brink.*
      - 11.1.5 *Fill the brink up to the welded bolt (halfway up brink).*
      - 11.1.6 *Once brink is "full," open valve R7 and close valve R1.*

11.1.7 *This will start your 50-minute cycle.*

11.2 *"Non-Hard-Pipes":*

11.2.1 *Close valve R6.*

11.2.2 *Close valve R4.*

11.2.3 *Water will start to fill the brink.*

11.2.4 *Add three gallons of Acid to the brink.*

11.2.5 *Fill the brink up to the welded bolt (halfway up brink).*

11.2.6 *Once brink is "full," open valve R7 and close valve R1.*

11.2.7 *This will start your 50-minute cycle.*

12. *After 10 minutes have elapsed, we need to bleed the 20µ filter housing and pinpoint carbonator (only on last run of week).*

12.1 *Open the front of the filter housing for a moment to let water flow out, creating contact with the pipe.*

12.2 *Open the dirty and clean drains (bottom of the filter housing) and let water flow for a moment.*

12.3 *Open the outlet of the filter housing and let water flow for a moment.*

12.4 *Open the top valve of the filter housing and let water flow for a moment.*

Ending the CFT Filler CIP cycle:

Caustic Only:

1. After the 50-minutes have elapsed, a hot flush needs to be performed.
2. Return to the CIP in the Cellar.
1. Open valve R1.
2. Close valve R7.
3. Open valve R6.
4. Open valve R4.
5. Let the hot-water cycle run for 20 minutes.
6. On the CIP screen on the HMI panel, press "Force" next to the "Empty Cycle On (Step 9)" label.

Note: This will start the draining cycle back to the brink.

7. Close the hot-water hose valve and valve R1.
8. Disconnect and neatly put away hot water hose.
9. Close the "Can Line Product Hose" and place a 2" cap on the end of the product line.
10. Neatly roll coil the "Can Line Product Hose."
11. Return to the CFT Filler.
12. From the HMI Screen, ensure the cycle has ended on the CIP screen.

13. Press the home screen button and “rolodex” up one page.
14. Turn all control buttons to the “Off” position.
15. “rolodex” up one screen and turn off the Machine Light.
16. Press the E-Stop button on the HMI and the electrical cabinet.
17. Open the doors of the filler to let air in.
18. Shut off the Air Supply to the filler.

After Acid Cycle:

1. On the CIP screen on the HMI panel, press “Force” next to the “Empty Cycle On (Step 9)” label.

Note: This will start the draining cycle back to the brink.

2. Disconnect and neatly put away cold water hose.
3. Close the “Can Line Product Hose” and place a 2” cap on the end of the product line.
4. Neatly roll coil the “Can Line Product Hose”.
5. Return to the CFT Filler.
6. From the HMI Screen, ensure the cycle has ended on the CIP screen.
7. Press the home screen button and “rolodex” up one page.
8. Turn all control buttons to the “Off” position.
9. “rolodex” up one screen and turn off the Machine Light.
10. Press the E-Stop button on the HMI and the electrical cabinet.
11. Open the doors of the filler to let air in.
12. Shut off the Air Supply to the filler.

Changeover Guide:

Filler:

1. For all Filler changeovers, the filler should not be running.
2. For all sizes, start with the following:
    - 2.1 From the HMI, select the bottle/can icon button.
    - 2.2 This screen will ask for a username and password: Use manu for both.
    - 2.3 Select the size you will be running, examples:
        - 2.3.1 12oz = 12pz on 1/25/2019 – Program 1.
        - 2.3.2 16oz = Stouty 16oz 5/19 – Program 7.
        - 2.3.3 19.2oz = 19.2oz Super Stouty – Program 2.
    - 2.4 Select the “Down arrow” button next to the “Floppy disk” button to download the program to the CFT Filler.
3. After the size has been selected, the CFT Filler bowl will need to be adjusted.

Note: When going from a higher size to a lower size can, change parts will need to be removed prior to lowering the CFT Filler bowl. Please see section #9 for more information

4. Select the “hand” button.
5. Select “Filler Height Adjustment”
6. Select “Filler Height Adjustment Enable/Disable” and ensure this is listed as “on”.
7. Turn the key on the HMI to jog mode.
8. Grab the jog wand and use the knob on the left of the wand labeled (increase/decrease) increase or decrease the filler height while watching the ruler guide to match up with the height setting listed on the side of the CFT Filler.
9. Once the height is set, see the following for the changeover parts.

Note: As stated in the earlier part of this guide, change parts will need to be removed instead of added when lowering the bowl to smaller sizes.

9.1 *16oz Can Height – Add in 16/19.2oz star wheel addition and add in 16oz guide rail.*
9.2 *19.2oz Can Height - Add in 16/19.2oz star wheel addition and add in 19.2oz guide rail.*
9.3 *12oz Can Height – Remove the 16/19.2oz star wheel and any added guide rails.*

10. *Once the parts are in/out and the bowl is set, disable the “Filler Height Adjustment Enable/Disable” button. The section should read as “off”.*
11. *Proceed to the Seamer Changeover for next steps.*

Seamer:

1. Open the Seamer door (next to lid slide) with an 8mm hex wrench.
2. On the big column, loosen the 10mm hex bolt. (Do not remove).
3. Close the Seamer door.
4. Open the Seamer door next to the outfeed of the Seamer using an 8mm hex wrench.
5. Move the jog wand to the outfeed of the Seamer.
6. Press reset on the jog wand to regain power to the machine while in jog mode.
7. Jog the seamer until the Seamer center column bolts are showing.
8. Loosen the two big columns (10mm hex wrench) like in step #2.
9. Loosen the 8mm and 6mm bolts from the center column using the corresponding hex wrenches.
10. Once the bolts have been loosened, jog the seamer so position “1” lines up with the first ops position indicator mark on the seamer table.
11. On the HMI, select the “hand” button.
12. Select “Seamer Height Adjustment.”

12.1 This will give you a reminder that the seamer block needs to be removed. This refers to the bolts that were loosened above.

13. "rolodex" up one page.
14. Select "Seamer Height Adjustment Enable/Disable" and ensure that it is in the "on" position.
15. Using the jog wand and corresponding "Seamer pin-height guides", raise or lower the seamer until the chuck barley touches the guide.
16. Proceed to increase the height by one click.

Note: The trainer will teach height adjustment.

17. For 19.2oz packaging, a separate star wheel will need to be added to the Seamer. Please refer to operator training on how to proceed.
18. Once the Seamer height is set, tighten down all the bolts that were loosened in sections # 2, #8 and #9.
19. Close the Seamer door and move the jog wand back to the front of the CFT Filler.
20. From the HMI screen, disable the "Seamer Height Adjustment Enable/Disable" button. This should read as "off."
21. The CFT Filler and Seamer should now be set to the corresponding height.

Deep Cleaning – Biofilm Remediation:

Biofilm-mediated bioremediation is an attractive approach for the elimination of environmental pollutants, because of its wide adaptability, biomass, and excellent capacity to absorb, immobilize, or degrade contaminants.

1. When it comes to Biofilm Remediation, an increase in strength of our foaming chemicals and the addition of a separate Acid is needed for our foaming units.
2. We would foam in the following order:
   - 2.1 F400 foaming.
     - 2.1.1 Rinse with warm water.
   - 2.2 FA6700 foaming.
     - 2.2.1 Rinse with cold water.
   - 2.3 Oxystrike Foaming.
     - 2.3.1 Rinse with cold water.
3. Please see below for chemical increase per Rochester Midland.
   - 3.1 Oxystrike Foaming Unit: 500ml of Oxystrike per 5 gallons of cold water along with normal Perafoam addition to Foaming Unit.
   - 3.2 F400 Foaming Unit: 12oz of F400 per gallon of warm water. (64oz per 5-gallon Foaming Unit).

3.3 FA6700 Foaming Unit: 8-10oz of FA6700 per gallon of cold water (40-50oz per 5-gallon Foaming Unit).
3.4 Perform foaming as normal while wearing proper PPE as outlined in this SOP.

Note: Do not let foam sit for longer than 10 minutes or allow it to dry.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

This guide is for “basic” troubleshooting of the CFT Can Filler and Seamer.

1. Filler In/out Fault:
   1.1 When a can is seen by the infeed eye and not seen by the outfeed filler eye, this fault will flag.
   1.2 Inspect the filler are for the missing or damaged can.
   1.3 Ensure this is removed from the filler area.
   1.4 Reset the fault and proceed to run.
2. Seamer In/out Fault:
   2.1 Like the Filler In/out fault, the seamer infeed eye detects a can entering the seamer when the outfeed detects the can leaving the seamer.
   2.2 When the fault is present, open the seamer doors and inspect the seamer area for loose or damaged cans. Remove any obstruction and verify that the seamer is cleared of any loose cans or lids.
   2.3 Reset fault and proceed to run.
3. Mouth seal issues:
   3.1 When cans are being filled, there may be times where beer is blowing past the lip of the can. This may indicate the mouth seal has failed but may not always be the cause.
   3.2 When this happens, stop the “Container Gate” and clear out the filler.
   3.3 Once the filler is clear of cans, stop the machine and jog to the problematic filling valve number.
   3.4 Remove the cup using the cup removal tool and inspect the mouth seal for wear, damage or if swollen.

3.5 If any of the above are present, replace the mouth seal by cleaning the cup, spraying the area with isopropyl alcohol and add in new mouth seal.

3.6 Insert the cup and mouth seal back into the filling valve area.

4. Crushing cans:

4.1 Cans being crush or dented at the lip may be an indication the filler height is set incorrectly (possibly too low or too high).

4.2 Readjust the filler height as needed.

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Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

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